

Research in Human- Computer Interaction (HCI)

Basanta Chapagain, Patan Multiple Campus

Research in HCI

- Research in the area of human-computer interaction (HCI) is **fascinating** and **complex**. It is **fascinating** because there are so **many interesting questions** and so many **changes over time** (due to **technical advancements**).
- It is complex because we borrow research methods from a number of different fields, modify them, and create our own “standards” for what is considered acceptable research.
- It is also complex because the research involves **human participants**, who naturally have **diverse behaviors, thoughts, and experiences**.
- It is important to understand the roots of the field, to understand the development of research methods in HCI, understand how HCI research has changed over time, and understand the multiple dimensions that must be considered when doing HCI research.

History of HCI Research

- The field of **Human-Computer Interaction (HCI)** is generally considered to have been formally established in **1982** with the first **Human Factors in Computing Systems conference** in Gaithersburg, Maryland, which later became the annual ACM SIGCHI conference. By 2026, HCI was about 44 **years old** as a recognized discipline.
- However, research related to HCI existed **before 1982**. Earlier work included projects like the **Social Security Administration project (1977)** involving task analysis, screen prototyping, and usability labs. Important contributions also came from early studies in **management, psychology, software engineering, and human factors**.
- For example, **Ben Shneiderman's book *Software Psychology* (1980)** and earlier ideas such as **Vannevar Bush's 1945 paper "As We May Think"** influenced the field. The **computer mouse**, demonstrated by **Doug Engelbart in 1968**, was another significant development.

History of HCI Research

- The rise of **personal computers in the late 1970s and early 1980s** (such as Apple II and IBM PC) played a major role in the growth of HCI. Computers were no longer limited to research labs and specialists; they began to be used in **homes and offices by non-technical users**. This shift created the need to focus on **usability and effective interaction between humans and computers**, leading to the development of HCI as a multidisciplinary field.

TYPES OF HCI RESEARCH CONTRIBUTIONS

- The field of **Human–Computer Interaction (HCI)** is **multidisciplinary**, drawing knowledge from fields such as computer science, sociology, psychology, communication, and various engineering disciplines.
- Many **research methods used in HCI originate from these fields**, but they are **adapted and modified** to suit HCI research. For example, methods like **experimental design and observation from psychology** are adjusted to study how people interact with computers.

Different types of research contributions:

- **Empirical contributions**—data (qualitative or quantitative) collected through any of the methods : experimental design, surveys, focus groups, time diaries, sensors and other automated means, ethnography, and other methods.

TYPES OF HCI RESEARCH CONTRIBUTIONS

- **Artifact contributions**—the design and development of new artifacts, including interfaces, toolkits, and architectures, mock-ups, and “envisionments.” These artifacts, are often accompanied by empirical data about feedback or usage. This type of contribution is often known as HCI systems research, HCI interaction techniques, or HCI design prototypes.
- **Methodological contributions**—new approaches that influence processes in research or practice, such as a new method, new application of a method, modification of a method, or a new metric or instrument for measurement.
- **Theoretical contributions**—concepts and models which are vehicles for thought, which may be predictive or descriptive, such as a framework, a design space, or a conceptual model.

TYPES OF HCI RESEARCH CONTRIBUTIONS

- **Dataset contributions**—a contribution which provides a corpus for the benefit of the research community, including a repository, benchmark tasks, and actual data.
- **Survey contributions**—a review and synthesis of work done in a specific area, to help identify trends and specific topics that need more work. This type of contribution can only occur after research in a certain area has existed for a few years so that there is sufficient work to analyze.
- **Opinion contributions**—writings which seek to persuade the readers to change their minds, often utilizing portions of the other contributions listed above, not simply to inform, but to persuade.

TYPES OF HCI RESEARCH CONTRIBUTIONS

Most **Human–Computer Interaction (HCI)** research falls into two main categories: **empirical research** and **artifact (system) contributions**.

- **Empirical research** focuses on studying **people or system use using data collection methods**
- **Artifact research** involves **developing systems or prototypes**.

Analysis of papers submitted to the **CHI 2016 conference** showed that **over 70% were empirical studies**, while about **28.4% were artifact or system papers**. Some studies combine both approaches, such as **building a prototype and then evaluating it with users**. Other types of research contributions are much less common.

CHANGES IN TOPICS OF HCI RESEARCH

- In the **1980s**, Human–Computer Interaction (HCI) research focused mainly on how people used **office automation software** such as word processors, databases, and statistical programs. Researchers studied interface elements like dialog boxes and error messages. Important early contributions included studies on **human error, training-based interface design, and direct manipulation**.
- By the **late 1980s and early 1990s**, **graphical user interfaces (GUIs)** and **usability engineering methods** became popular.
- In the **1990s**, the rise of the **Internet and the web** expanded HCI research to include **web pages, email, instant messaging, and groupware**, bringing communication studies into HCI.

CHANGES IN TOPICS OF HCI RESEARCH

- From **2004 onward**, research shifted toward **user-generated content** such as blogs, photos, videos, and social networking. Researchers also began focusing more on **diverse users**, including children, older adults, and people with disabilities. Interest in **touchscreens and multitouch interaction** increased in the late 2000s.
- By the **late 2010s**, HCI research expanded beyond simple task performance to topics like **collaboration, user experience, emotions, and communication**.
- Modern research areas include **mobile devices, gestures, wearable and embedded computing, sensors, sustainability, big data, social computing, and accessibility, Machine Learning and AI**, and these topics continue to evolve with technological and societal changes.

CHANGES IN HCI RESEARCH METHODS

- Research methods in **Human–Computer Interaction (HCI)** have evolved due to technological advancements, new tools, and changing research opportunities. Previously expensive tools such as eye-tracking, sensors, **EEG**, and **EMG** are now more affordable, allowing more researchers to use them. New platforms and tools like **Amazon Mechanical Turk** enable large-scale online data collection. The rise of social media, health tracking, and digital records has also created massive datasets that researchers can analyze.
- Unlike fields such as sociology or economics, HCI researchers usually **collect their own data**, which traditionally led to **smaller datasets** (around 50–100 participants). Recently, **big data approaches** allow studies involving tens of thousands of users. However, big data often shows **correlations rather than causes**, so combining large datasets with **small interactive studies** (interviews, focus groups) helps researchers better understand results.

CHANGES IN HCI RESEARCH METHODS

- Another limitation in HCI is the **lack of longitudinal studies**, mainly because technology changes rapidly. Despite this, some long-term studies—such as research on internet use and social networking—have provided valuable insights.
- HCI research is also complex because **participants must represent the actual users of a system** (e.g., lawyers for legal software), making recruitment difficult and time-consuming. Additionally, researchers in HCI come from many academic backgrounds, such as computer science, psychology, sociology, and engineering. As a result, HCI uses a **wide range of research methods from multiple disciplines** to study how people interact with technology.

Understanding HCI Research Methods and Measurement

- HCI research must balance **rigorous scientific methods** with **practical relevance**. The goal is to produce results that improve **interface design, user training, development processes, and policies**, while maintaining strong research standards. Because HCI focuses on **human-centered computing**, researchers must support their ideas with solid evidence and evaluation.
- In early HCI research, measurement focused on **human performance metrics** from psychology and human factors, such as **task completion time, error rate, task success, learning time, retention, and user satisfaction**. These quantitative measures are still widely used, especially when tasks can be clearly defined and measured.

Understanding HCI Research Methods and Measurement

- However, modern HCI research also studies broader human experiences such as **motivation, collaboration, trust, emotions, and social participation**. These aspects are harder to measure using traditional laboratory experiments. To understand them, researchers often use **multiple research methods**, including **interviews, observations, case studies, data logging, and longitudinal studies**.
- Researchers like Ben Shneiderman describe two perspectives: **micro-HCI**, which focuses on task performance and usability metrics, and **macro-HCI**, which studies larger social and emotional impacts of technology. Because modern technologies involve communication, collaboration, and social interaction, HCI increasingly adopts **research methods from social sciences** to better understand why and how people use technology.

Research in the Field of Computer Applications

- Research in **Computer Applications** focuses on developing and improving software systems and technologies that help solve real-world problems. It involves studying how computers and software can be applied in different fields such as **business, education, healthcare, communication, and entertainment**.
- The main goal of research in computer applications is to **design efficient, user-friendly, and reliable systems**. Researchers work on areas such as **software development, databases, web technologies, artificial intelligence, data analysis, cybersecurity, and human-computer interaction**. These studies help improve system performance, security, usability, and the overall user experience.
- Research in this field often uses different methods such as **experiments, system development, data analysis, and user testing**. The results of this research help create **new applications, improve existing technologies, and support innovation in the digital world**.

Science and the Field of Computer Applications

- The **field of computer applications** is deeply connected to **science**, as it applies scientific principles to **design, develop, and improve software and computing systems**.
- Science provides the **foundation of logical reasoning, experimentation, measurement, and problem-solving**, which is essential for creating effective computer applications.

Connection of Science and Field of Computer Applications

Principles and Theory:

- Computer applications rely on concepts from **mathematics, physics, and logic**, which are scientific foundations.
- Algorithms, data structures, and software architecture are based on **scientific and mathematical reasoning**.

Science and the Field of Computer Applications

Experimentation and Testing:

- Developing applications involves systematic testing and experimentation, similar to scientific experiments.
- This ensures accuracy, reliability, and efficiency of programs.

Data and Analysis:

- Computer applications often collect and process large amounts of data.
- Scientific methods guide how this data is analyzed, interpreted, and used to make decisions.

Science and the Field of Computer Applications

Problem-Solving:

- Science emphasizes understanding problems and finding solutions using evidence and logic.
- Computer applications apply this approach to real-world tasks, such as automation, simulation, communication, and information management.

Innovation and Research:

- Just as science explores **new phenomena**, research in computer applications explores **new technologies, software tools, and methods**.
- This includes areas like **artificial intelligence, data analytics, and human-computer interaction**.

Science and the Field of Computer Applications

- In Summary , Science provides the **methodological rigor and logical foundation**, while the **field of computer applications applies these principles** to develop technologies that solve practical problems, improve productivity, and enhance daily life.